MD - 216 - 49EA - EG - CV - DED - 1001
DRAWING NUMBER
REV D01

SCHEDULE OF STORM WATER DRAINAGE SYSTEM

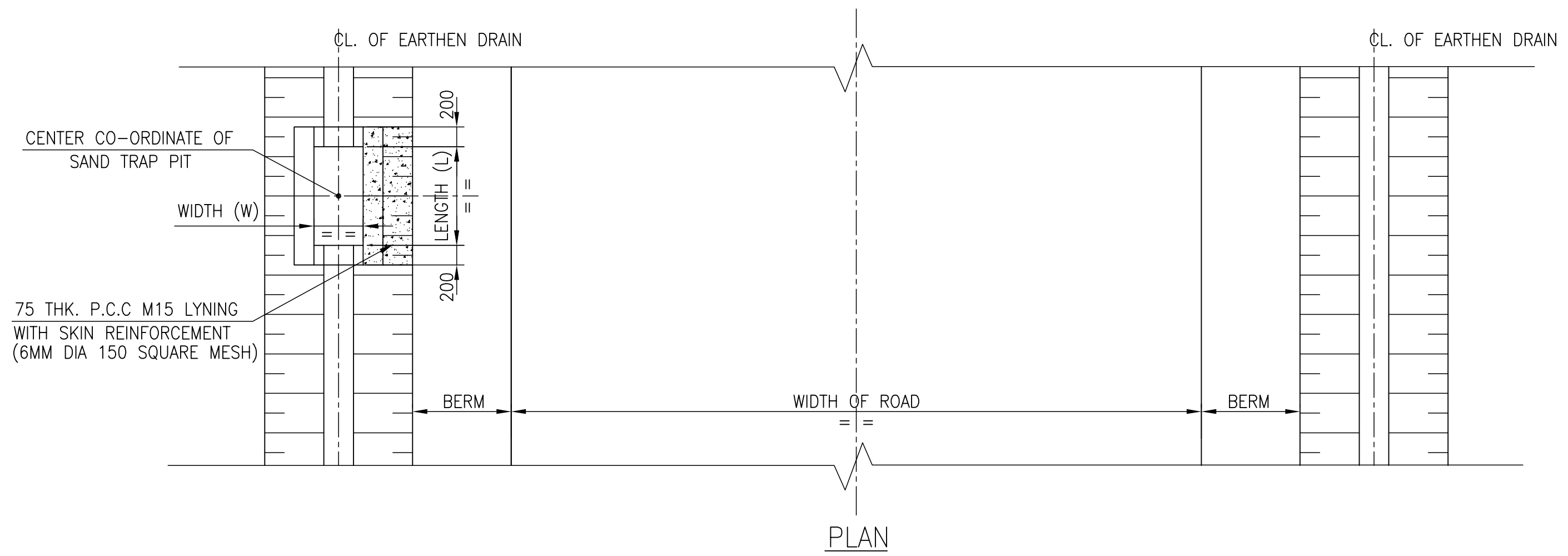
SL. NO.	NODE		DRAIN MKD.	LENGTH (M)	CL. OF CO-ORDINATE (M)		INVERT LEVELS (M)		SLOPE (%)	DRAIN TYPE	SHAPE	REMARKS
	START	END			START NODE	END NODE	START NODE	END NODE				
1	1	2	D1	95.58	E=263668.975 N=2506278.233	E=263744.793 N=2506272.428	914.700	914.510	0.201	EARTHEN TRAPEZOIDAL	300	
2	3	5	D2	46.48	E=263697.649 N=2506284.359	E=263728.544 N=2506260.839	915.150	915.040	0.218	EARTHEN TRAPEZOIDAL	300	
3	4	5a	D3	28.88	E=263694.676 N=2506237.103	E=263718.421 N=2506251.941	915.200	915.080	0.411	EARTHEN TRAPEZOIDAL	300	
4	5a	6b	D3a	10	E=263718.421 N=2506251.941	E=263723.720 N=2506243.460	915.080	914.660	4.181	R.C. PIPE DRAIN	ø300	
5	6	6a	D4	4.24	E=263711.000 N=2506235.511	E=263714.599 N=2506237.761	914.710	914.700	0.200	EARTHEN TRAPEZOIDAL	300	
6	6a	6b	D4a	11.63	E=263714.599 N=2506237.761	E=263723.720 N=2506243.460	914.700	914.660	0.310	R.C. PIPE DRAIN	ø300	
7	6b	7	D4b	12.43	E=263723.720 N=2506243.460	E=263734.999 N=2506250.508	914.660	914.630	0.307	R.C. PIPE DRAIN	ø300	
8	7	7a	D4c	13.52	E=263734.999 N=2506250.508	E=263746.467 N=2506257.674	914.630	914.490	0.992	R.C. PIPE DRAIN	ø400	
9	5	7	D5	12.18	E=263728.544 N=2506260.839	E=263734.999 N=2506250.508	915.040	914.530	4.260	R.C. PIPE DRAIN	ø300	
10	2	8	D5a	12.47	E=263744.793 N=2506272.428	E=263751.894 N=2506261.065	914.510	914.480	0.235	R.C. PIPE DRAIN	ø500	
11	7a	8	D6	6.4	E=263746.467 N=2506257.674	E=263751.894 N=2506261.065	914.490	914.480	0.200	EARTHEN TRAPEZOIDAL	300	
12	8	10	D7	59.63	E=263751.894 N=2506261.065	E=263776.951 N=2506226.068	914.480	914.360	0.201	EARTHEN TRAPEZOIDAL	300	
13	9	9T	D8	42.5	E=263736.669 N=2506200.897	E=263772.711 N=2506223.418	914.750	914.560	0.452	EARTHEN TRAPEZOIDAL	300	
14	9T	10	D8T	5	E=263772.711 N=2506223.418	E=263776.951 N=2506226.068	914.560	914.360	3.999	EARTHEN TRAPEZOIDAL	300	TRANSITION
15	10	12	D9	9.93	E=263776.951 N=2506226.068	E=263782.250 N=2506217.587	914.360	914.330	0.282	R.C. PIPE DRAIN	ø500	
16	11ii	11iiT	D10	14.48	E=263765.768 N=2506207.289	E=263778.010 N=2506214.938	914.500	914.370	0.932	EARTHEN TRAPEZOIDAL	300	
17	11iiT	12	D10T	5.01	E=263778.010 N=2506214.938	E=263782.250 N=2506217.587	914.370	914.330	0.698	EARTHEN TRAPEZOIDAL	300	TRANSITION
18	12	13	D11	21.65	E=263782.250 N=2506217.587	E=263798.950 N=2506211.622	914.330	914.290	0.208	EARTHEN TRAPEZOIDAL	300	
19	13	13i	D12	13.4	E=263798.950 N=2506211.622	E=263806.049 N=2506200.259	914.290	914.020	1.978	R.C. PIPE DRAIN	ø500	
20	13i	13a	D13	24.77	E=263806.049 N=2506200.259	E=263819.159 N=2506179.851	914.020	913.960	0.242	EARTHEN TRAPEZOIDAL	500	
21	13a	64b	D13y	15.39	E=263819.159 N=2506179.851	E=263832.214 N=2506188.009	913.960	913.800	1.039	R.C. PIPE DRAIN	ø600	
22	14	14i	D13a	17.09	E=263800.595 N=2506147.465	E=263809.653 N=2506132.969	913.900	913.520	2.223	R.C. PIPE DRAIN	ø300	
23	14i	15i	D13b	116	E=263809.653 N=2506132.969	E=263879.113 N=2506062.573	913.520	913.060	0.397	EARTHEN TRAPEZOIDAL	300	
24	14i	15i	D13c	148.26	E=263809.653 N=2506132.969	E=263879.113 N=2506062.573	913.520	913.060	0.310	EARTHEN TRAPEZOIDAL	300	
25	15i	15	D13d	15.79	E=263879.113 N=2506062.573	E=263887.539 N=2506049.090	913.060	912.970	0.570	R.C. PIPE DRAIN	ø500	
26	15	17	D14	126.18	E=263887.539 N=2506049.090	E=263940.430 N=2505955.011	912.900	912.520	0.301	EARTHEN TRAPEZOIDAL	500	
27	11i	11iT	D15	188.61	E=263731.953 N=2506182.983	E=263817.873 N=2506020.320	914.500	913.170	0.705	EARTHEN TRAPEZOIDAL	300	
28	11iT	16	D15T	4.62	E=263817.873 N=2506020.320	E=263820.295 N=2506016.445	913.170	913.140	0.758	EARTHEN TRAPEZOIDAL	300	TRANSITION
29	16	16T	D16	172.91	E=263820.295 N=2506016.445	E=263936.162 N=2505952.344	913.140	912.540	0.344	EARTHEN TRAPEZOIDAL	300	
30	16T	17	D16T	5.03	E=263936.162 N=2505952.344	E=263940.430 N=2505955.011	912.540	912.520	0.397	EARTHEN TRAPEZOIDAL	300	TRANSITION
31	17	19	D17	10	E=263940.430 N=2505955.011	E=263945.730 N=2505946.531	912.520	912.460	0.600	R.C. PIPE DRAIN	ø500	
32	18	18T	D18	45.34	E=263901.609 N=2505918.961	E=263941.461 N=2505943.863	912.700	912.490	0.463	EARTHEN TRAPEZOIDAL	300	
33	18T	19	D18T	6.69	E=263941.461 N=2505943.863	E=263945.730 N=2505946.531	912.490	912.460	0.448	EARTHEN TRAPEZOIDAL	300	TRANSITION
34	19	19i	D19	94.11	E=263945.730 N=2505946.531	E=263997.621 N=2505883.145	912.460	912.180	0.298	EARTHEN TRAPEZOIDAL	750	
35	19i	20i	D19a	13.42	E=263997.621 N=2505883.145	E=264004.734 N=2505871.761	912.180	912.100	0.596	R.C. PIPE DRAIN	ø500	
36	20i	20	D19b	44.91	E=264004.734 N=2505871.761	E=264013.169 N=2505838.605	912.100	912.010	0.200	EARTHEN TRAPEZOIDAL	750	
37	18	18i	D20	87.73	E=263901.609 N=2505918.961	E=263935.832 N=2505844.535	912.690	912.520	0.197	EARTHEN TRAPEZOIDAL	300	
38	18i	18ii	D20a	13.42	E=263935.832 N=2505844.535	E=263942.945 N=2505833.152	912.520	912.480	0.298	R.C. PIPE DRAIN	ø400	
39	18ii	18iiT	D20b	97.47	E=263901.609 N=2505918.961	E=264008.893 N=2505835.933	912.480	912.050	0.464	EARTHEN TRAPEZOIDAL	300	
40	18iiT	20	D20bT	5.08	E=264008.893 N=2505835.933	E=264013.169 N=2505838.605	912.050	912.010	0.709	EARTHEN TRAPEZOIDAL	300	TRANSITION
41	20	22	D21	10.43	E=264013.169 N=2505838.605	E=264018.468 N=2505830.124	912.010	911.970	0.384	R.C. PIPE DRAIN	ø600	
42	21	21T	D22	46.92	E=263974.348 N=2505802.555	E=264014.193 N=2505827.452	912.600	912.050	1.164	EARTHEN TRAPEZOIDAL	300	
43	21T	22	D22T	5.1	E=264014.193 N=2505827.452	E=264018.468 N=2505830.124	912.050	911.970	1.646	EARTHEN TRAPEZOIDAL	300	TRANSITION
44	22	23	D23	25.58	E=264018.468 N=2505830.124	E=264040.746 N=2505827.119	911.970	911.940	0.137	EARTHEN TRAPEZOIDAL	800	
45	23	24	D24	13.4	E=264040.746 N=2505827.119	E=264052.110 N=2505834.220	911.940	911.880	0.410	R.C. PIPE DRAIN	ø600	
46	25	25i	D25	53.95	E=264123.850 N=2505808.200	E=264169.604 N=2505836.790	912.100	911.990	0.204	EARTHEN TRAPEZOIDAL	300	
47	25i	32	D26	25.97	E=264169.604 N=2505836.790	E=264191.629 N=2505850.553	911.990	911.640	1.348	R.C. PIPE DRAIN	ø400	
48	26	26T	D27	121.74	E=263992.592 N=2505946.453	E=264049.433 N=2505838.503	912.520	911.960	0.463	EARTHEN TRAPEZOIDAL	300	
49	26T	24	D27T	5.06	E=264049.433 N=2505838.503	E=264052.110 N=2505834.220	911.960	911.880	1.581	EARTHEN TRAPEZOIDAL	300	TRANSITION
50	24	31	D27a	151.4	E=264052.110 N=2505834.220	E=264185.058 N=2505861.069	911.880	911.680	0.132	EARTHEN TRAPEZOIDAL	800	
51	31	31i	D27b	13.85	E=264185.058 N=2505861.069	E=264187.921 N=2505873.470	911.680	911.640	0.289	EARTHEN TRAPEZOIDAL	800	
52	26	26i	D28	117.97	E=263992.592 N=2505946.453	E=264092.640 N=2506008.969	912.550	912.320	0.199	EARTHEN TRAPEZOIDAL	300	
53	26i	28	D28a	78.63	E=264092.640 N=2506008.969	E=264128.086 N=2505952.243	912.320	912.210	0.131	EARTHEN TRAPEZOIDAL	300	
54	27	27T	D29	12.06	E=264113.068 N=2505942.859	E=264125.471 N=2505950.609	912.550	912.280	2.206	EARTHEN TRAPEZOIDAL	300	
55	27T	28	D29T	3.02	E=264125.471 N=2505950.609	E=264128.086 N=2505952.243	912.280	912.210	2.381	EARTHEN TRAPEZOIDAL	300	TRANSITION
56	28	30	D30	9.04	E=264128.086 N=2505952.243	E=264133.703 N=2505943.254	912.210	912.190	0.299	R.C. PIPE DRAIN	ø400	
57	29	29T	D31	11.47	E=264118.685 N=2505933.869	E=264130.607 N=2505941.319	912.350	912.220	1.134	EARTHEN TRAPEZOIDAL	300	
58	29T	30	D31T	3.51	E=264130.607 N=2505941.319	E=264133.703 N=2505943.254	912.220	912.190	0.997	EARTHEN TRAPEZOIDAL	300	TRANSITION
59	30	30T	D32	88.17	E=264133.703 N=2505943.254	E=264185.505 N=2505877.337	912.190	911.670	0.586	EARTHEN TRAPEZOIDAL	300	
60	30T	31i	D32T	4.98	E=264185.505 N=2505877.337	E=264187.921 N=2505873.470	911.670	911.640	0.562	EARTHEN TRAPEZOIDAL	300	TRANSITION
61	31i	41i	D33	12.54	E=264187.921 N=2505873.470	E=264198.437 N=2505880.041	911.640	911.600	0.319	R.C. PIPE DRAIN	ø800	
62	33	33T	D34	95.54	E=263923.981 N=2506090.124	E=263997.765 N=2506098.527	912.800	912.610	0.199	EARTHEN TRAPEZOIDAL	300	
63	33T	33i	D34T	5.01	E=263997.765 N=2506098.527	E=264000.342 N=2506094.402	912.610	912.600	0.200	EARTHEN TRAPEZOIDAL	300	TRANSITION
64	33i	37	D34a	133.59	E=264000.342 N=2506094.402	E=264088.976 N=2506031.816	912.600	912.350	0.187	EARTHEN TRAPEZOIDAL	300	
65	34	35	D35	123.42	E=263910.852 N=2506060.280	E=263975.580 N=2505956.693	912.800	912.550	0.203	EARTHEN TRAPEZOIDAL	300	
66	35	36	D36	9.62	E=263975.580 N=2505956.693	E=263986.807 N=2505963.709	912.550	912.520	0.312	R.C. PIPE DRAIN	ø500	
67	36	37	D37	131.39	E=263986.807 N=2505963.709	E=264088.976 N=2506031.816	912.520	912.350	0.129	EARTHEN TRAPEZOIDAL	300	
68	37	39	D38	12.39	E=264088.976 N=2506031.816	E=264099.448 N=2506038.457	912.350	912.310	0.299	R.C. PIPE DRAIN	ø600	
69	38	38T	D39	173.2	E=264099.158 N=2506055.904	E=264212.461 N=2505999.509	912.600	912.130	0.271	EARTHEN TRAPEZOIDAL	300	
70	38T	40	D39T	7.36	E=264212.461 N=2505999.509	E=264208.390 N=2505993.566	912.130	912.100	0.408	EARTHEN TRAPEZOIDAL	300	TRANSITION
71	38	39T	D40	10.4	E=264099.158 N=2506055.904	E=264096.966 N=2506042.430	912.530	912.370	1.561	EARTHEN TRAPEZOIDAL	300	
72	39T	39	D40T	4.9	E=264096.966 N=2506042.430	E=264099.448 N=2506038.457	912.370	912.310	1.122	EARTHEN TRAPEZOIDAL	300	TRANSITION

SCHEDULE OF STORM WATER DRAINAGE SYSTEM

SL. NO.	NODE		DRAIN MKD.	LENGTH (M)	CL. OF CO-ORDINATE (M)		INVERT LEVELS (M)		SLOPE (%)	DRAIN TYPE	SHAPE	REMARKS
	START	END			START NODE	END NODE	START NODE	END NODE				
73	39	40	D41	160.51	E=264099.448 N=2506038.457	E=264208.390 N=2505993.566	912.310	912.100	0.133	EARTHEN TRAPEZOIDAL	300	
74	40	42	D42	12.69	E=264208.390 N=2505993.566	E=264214.961 N=2505983.050	912.100	912.030	0.591	R.C. PIPE DRAIN	ø600	
75	41	42T	D43	46.1	E=264164.418 N=2505951.467	E=264210.223 N=2505980.089	912.200	912.040	0.338	EARTHEN TRAPEZOIDAL	300	
76	42T	42	D43T	5.38	E=264210.223 N=2505980.089	E=264214.961 N=2505983.050	912.040	912.030	0.353	EARTHEN TRAPEZOIDAL	300	TRANSITION
77	42	43T	D44	92.76	E=264214.961 N=2505983.050	E=264264.748 N=2505912.544	912.030	911.560	0.501	EARTHEN TRAPEZOIDAL	500	
78	43T	43	D44T	5.14	E=264264.748 N=2505912.544	E=264261.382 N=2505908.761	911.560	911.450	2.140	EARTHEN TRAPEZOIDAL	500	TRANSITION
79	41	41T	D45	74.57	E=264164.418 N=2505951.467	E=264195.866 N=2505884156	911.980	911.630	0.480	EARTHEN TRAPEZOIDAL	300	
80	41T	41i	D45T	5.07	E=264195.866 N=2505884156	E=264198.437 N=2505880.041	911.630	911.600	0.513	EARTHEN TRAPEZOIDAL	300	TRANSITION
81	41i	43	D45a	73.53	E=264198.437 N=2505880.041	E=264261.382 N=2505908.761	911.600	911.450	0.204	EARTHEN TRAPEZOIDAL	800	
82	43	44	D46	12.4	E=264261.382 N=2505908.761	E=264267.953 N=2505898.246	911.450	911.400	0.403	R.C. PIPE DRAIN	ø800	
83	32	44	D47	90.0	E=264191.629 N=2505850.553	E=264267.953 N=2505898.246	911.640	911.400	0.267	EARTHEN TRAPEZOIDAL	500	
84	44	45	D48	29.38	E=264267.953 N=2505898.246	E=264292.843 N=2505913.798	911.400	911.200	0.681	R.C. PIPE DRAIN	ø800	
85	45	45T	D49	21.55	E=264292.843 N=2505913.798	E=264311.141 N=2505925.233	911.200	911.130	0.339	EARTHEN TRAPEZOIDAL	800	
86	45T	50	D49T	5.08	E=264311.141 N=2505925.233	E=264315.444 N=2505927.921	911.130	911.100	0.532	EARTHEN TRAPEZOIDAL	800	TRANSITION
87	46i	47	D50	110.3	E=264197.193 N=2506117.163	E=264260.453 N=2506032.909	912.400	911.960	0.399	EARTHEN TRAPEZOIDAL	300	
88	46	48	D51	110.2	E=264175.483 N=2506103.597	E=264223.478 N=2506009.804	912.400	911.960	0.399	EARTHEN TRAPEZOIDAL	300	
89	48	48T	D52	131.53	E=264223.478 N=2506009.804	E=264304.842 N=2505935.918	911.960	911.440	0.394	EARTHEN TRAPEZOIDAL	300	
90	48T	49	D52T	4.66	E=264304.842 N=2505935.918	E=264308.873 N=2505938.437	911.440	911.420	0.564	EARTHEN TRAPEZOIDAL	300	TRANSITION
91	47	47T	D53	105.93	E=264260.665 N=2506032.569	E=264312.186 N=2505942.110	911.960	911.440	0.492	EARTHEN TRAPEZOIDAL	300	
92	47T	49	D53T	5.17	E=264312.186 N=2505942.110	E=264308.873 N=2505938.437	911.440	911.420	0.464	EARTHEN TRAPEZOIDAL	300	TRANSITION
93	49	50	D54	12.3	E=264308.873 N=2505938.437	E=264315.444 N=2505927.921	911.420	911.100	2.562	R.C. PIPE DRAIN	ø500	
94	50	53	D55	52.7	E=264315.444 N=2505927.921	E=264360.136 N=2505955.848	911.100	910.950	0.285	EARTHEN TRAPEZOIDAL	1000	
95	63	63i	D56	32.89	E=264279.593 N=2506113.617	E=264297.019 N=2506085.728	911.750	911.690	0.198	EARTHEN TRAPEZOIDAL	500	
96	63i	63iT	D56a	131.49	E=264297.019 N=2506085.728	E=264359.195 N=2505972.602	911.690	911.420	0.200	EARTHEN TRAPEZOIDAL	500	
97	63iT	52	D56aT	8.6	E=264359.195 N=2505972.602	E=264353.565 N=2505966.364	911.420	911.400	0.314	EARTHEN TRAPEZOIDAL	500	TRANSITION
98	60	51	D57	82.46	E=264228.880 N=2506106.836	E=264272.579 N=2506036.903	911.680	911.570	0.133	EARTHEN TRAPEZOIDAL	500	
99	51	52	D57a	130.6	E=264272.579 N=2506036.903	E=264353.565 N=2505966.364	911.570	911.400	0.134	EARTHEN TRAPEZOIDAL	500	
100	52	53	D58	12.4	E=264353.565 N=2505966.364	E=264360.136 N=2505955.848	911.400	911.320	0.605	R.C. PIPE DRAIN	ø500	
101	54	54T	D59	100.39	E=264063.824 N=2506177.508	E=264121.944 N=2506101.480	912.700	912.350	0.349	EARTHEN TRAPEZOIDAL	300	
102	54T	55	D59T	5	E=264121.944 N=2506101.480	E=264124.687 N=2506097.091	912.350	912.290	1.201	EARTHEN TRAPEZOIDAL	300	TRANSITION
103	54i	54iT	D60	70.53	E=264051.103 N=2506169.560	E=263994.739 N=2506129.359	912.700	912.560	0.199	EARTHEN TRAPEZOIDAL	300	
104	54iT	55i	D60T	5.04	E=263994.739 N=2506129.359	E=263996.106 N=2506124.581	912.560	912.550	0.198	EARTHEN TRAPEZOIDAL	300	TRANSITION
105	55i	55	D60a	196.82	E=263996.106 N=2506124.581	E=264124.687 N=2506097.091	912.550	912.290	0.132	EARTHEN TRAPEZOIDAL	300	
106	55	59	D61	12.3	E=264124.687 N=2506097.091	E=264135.203 N=2506103.662	912.290	912.260	0.285	R.C. PIPE DRAIN	ø500	
107	56	57	D62	77.65	E=264089.605 N=2506193.617	E=264123.805 N=2506121.902	912.700	912.450	0.322	EARTHEN TRAPEZOIDAL	300	
108	57	58	D63	12.2	E=264123.805 N=2506121.902	E=264130.376 N=2506111.387	912.450	912.420	0.287	R.C. PIPE DRAIN	ø400	
109	58	59	D64	9.27	E=264130.376 N=2506111.387	E=264135.203 N=2506103.662	912.420	912.260	1.727	EARTHEN TRAPEZOIDAL	300	
110	59	59T	D65	113.35	E=264135.203 N=2506103.662	E=264226.189 N=2506111.142	912.260	911.900	0.313	EARTHEN TRAPEZOIDAL	300	
111	59T	60	D65T	5.05	E=264226.189 N=2506111.142	E=264228.880 N=2506106.836	911.900	911.680	4.355	EARTHEN TRAPEZOIDAL	300	TRANSITION
112	60	63	D66	VOID								
113	61i	61	D67	61.43	E=264113.664 N=2506208.652	E=264165.760 N=2506241.204	912.635	912.510	0.203	EARTHEN TRAPEZOIDAL	300	
114	61	62	D68	21.0	E=264165.760 N=2506241.204	E=264183.568 N=2506252.333	912.510	912.460	0.240	R.C. PIPE DRAIN	ø400	
115	62	63	D69	233.64	E=264183.568 N=2506252.333	E=264279.593 N=2506113.617	912.460	911.750	0.305	EARTHEN TRAPEZOIDAL	500	
116	64i	64	D70	VOID								
117	64b	65	D71	86.43	E=263832.214 N=2506188.009	E=263909.630 N=2506174.195	913.800	913.570	0.266	EARTHEN TRAPEZOIDAL	750	
118	64	64T	D71a	31.57	E=263812.036 N=2506218.418	E=263828.806 N=2506191.725	914.140	913.830	0.982	EARTHEN TRAPEZOIDAL	300	
119	64T	64b	D71T	5.04	E=263828.806 N=2506191.725	E=263832.214 N=2506188.009	913.830	913.800	0.595	EARTHEN TRAPEZOIDAL	300	TRANSITION
120	65	65i	D72	42.48	E=263909.630 N=2506174.195	E=263945.657 N=2506196.707	913.570	912.980	1.401	EARTHEN TRAPEZOIDAL	750	
121	65i	66	D72a	139.46	E=263945.657 N=2506196.707	E=264060.314 N=2506268.352	912.980	912.390	0.419	EARTHEN TRAPEZOIDAL	750	
122	66	67	D73	13.97	E=264060.314 N=2506268.352	E=264072.187 N=2506275.771	912.390	912.320	0.537	R.C. PIPE DRAIN	ø600	
123	67	67T	D74	41.3	E=264072.187 N=2506275.771	E=264109.552 N=2506272.639	912.320	912.120	0.472	EARTHEN TRAPEZOIDAL	750	
124	67T	67i	D74T	4.04	E=264109.552 N=2506272.639	E=264113.879 N=2506270.570	912.120	912.100	0.496	EARTHEN TRAPEZOIDAL	750	TRANSITION
125	67i	82	D74a	VOID								
126	68	69	D75	23.03	E=2638957.334 N=2506150.889	E=263915.263 N=2506163.092	913.200	912.970	1.017	EARTHEN TRAPEZOIDAL	300	
127	69	69T	D76	29.89	E=263915.263 N=2506163.092	E=263940.615 N=2506178.933	912.970	912.860	0.375	EARTHEN TRAPEZOIDAL	300	
128	69T	69a	D76T	6.16	E=263940.615 N=2506178.933	E=263945.721 N=2506182.125	912.860	912.850	0.211	EARTHEN TRAPEZOIDAL	300	TRANSITION
129	69a	69b	D76B	30.22	E=263945.721 N=2506182.125	E=263971.538 N=2506198.256	912.850	912.680	0.571	R.C. PIPE DRAIN	ø300	
130	69b	69i	D76C	13.89	E=263971.538 N=2506198.256	E=263983.172 N=2506205.527	912.680	912.650	0.198	EARTHEN TRAPEZOIDAL	300	
131	69i	69iT	D76D	7.4	E=263983.172 N=2506205.527	E=263989.523 N=2506209.495	912.650	912.640	0.192	EARTHEN TRAPEZOIDAL	300	
132	69iT	69ii	D76DT	3.02	E=263989.523 N=2506209.495	E=263992.103 N=2506211.107	912.640	912.630	0.259	EARTHEN TRAPEZOIDAL	300	TRANSITION
133	69ii	69iii	D76E	23.68	E=263992.103 N=2506211.107	E=264012.015 N=2506223.550	912.630	912.560	0.300	R.C. PIPE DRAIN	ø500	
134	69iii	69iv	D76F	55.02	E=264012.015 N=2506223.550	E=264058.904 N=2506252.850	912.560	912.450	0.200	EARTHEN TRAPEZOIDAL	300	
135	69iv	69v	D76G	18.34	E=264058.904 N=2506252.850	E=264074.401 N=2506262.533	912.450	912.390	0.300	R.C. PIPE DRAIN	ø500	
136	69v	81i	D76a	26.55	E=264074.401 N=2506262.533	E=264101.546 N=2506262.864	912.390	912.180	0.798	EARTHEN TRAPEZOIDAL	500	
137	68	70	D77	118.93	E=2638957.334 N=2506150.889	E=263936.811 N=2506123.375	913.200	912.810	0.328	EARTHEN TRAPEZOIDAL	300	
138	70	70T	D78	49.57	E=263936.811 N=2506123.375	E=263978.077 N=2506143.934	912.810	912.700	0.222	EARTHEN TRAPEZOIDAL	300	
139	70T	71	D78T	6.04	E=263978.077 N=2506143.934	E=263977.405 N=2506149.685	912.700	912.690	0.248	EARTHEN TRAPEZOIDAL	300	TRANSITION
140	71	72	D79	10.53	E=263977.405 N=2506149.685	E=263986.676 N=2506155.478	912.690	912.640	0.475	R.C. PIPE DRAIN	ø300	
141	72	73	D80	16.71	E=263986.676 N=2506155.478	E=264002.082 N=2506153.549	912.640	912.610	0.150	EARTHEN TRAPEZOIDAL	300	
142	73	74	D81	20.64	E=264002.082 N=2506153.549	E=264019.737 N=2506164.823	912.610	912.550	0.291	R.C. PIPE DRAIN	ø400	
143	74	75	D82	15.4	E=264019.737 N=2506164.823	E=264022.971 N=2506178.158	912.550	912.530	0.162	EARTHEN TRAPEZOIDAL	300	
144	75	76	D83	21.56	E=264022.971 N=2506178.158	E=264041.269 N=2506189.592	912.530	912.460	0.302	R.C. PIPE DRAIN	ø400	

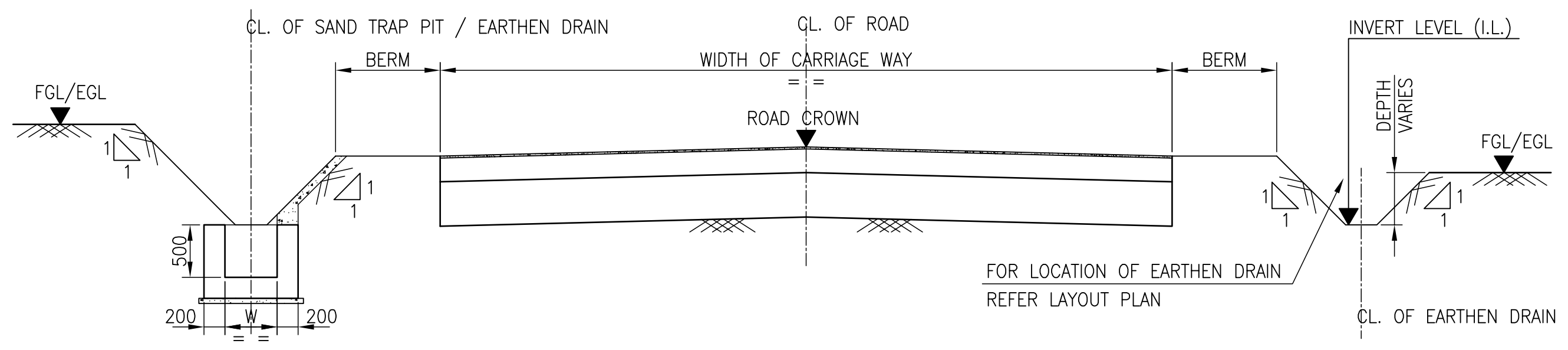
SCHEDULE OF STORM WATER DRAINAGE SYSTEM

SL. NO.	NODE		DRAIN MKD.	LENGTH (M)	CL. OF CO-ORDINATE (M)		INVERT LEVELS (M)		SLOPE (%)	DRAIN TYPE	SHAPE	REMARKS
	START	END			START NODE	END NODE	START NODE	END NODE				
145	76	77	D84	21.94	E=264041.269 N=2506189.592	E=264053.703 N=2506190.876	912.460	912.430	0.137	EARTHEN TRAPEZOIDAL	300	
146	77	78	D85	20.32	E=264053.703 N=2506190.876	E=264071.003 N=2506201.686	912.430	912.370	0.295	R.C. PIPE DRAIN	ø500	
147	78	79	D86	32.06	E=264071.003 N=2506201.686	E=264089.130 N=2506219.499	912.370	912.330	0.140	EARTHEN TRAPEZOIDAL	300	
148	79	80	D87	10.5	E=264089.130 N=2506219.499	E=264097.946 N=2506225.007	912.330	912.300	0.286	R.C. PIPE DRAIN	ø500	
149	80	81	D88	32.6	E=264097.946 N=2506225.007	E=264118.004 N=2506236.571	912.300	912.250	0.138	EARTHEN TRAPEZOIDAL	300	
150	81	81T	D88a	26.08	E=264118.004 N=2506236.571	E=264104.264 N=2506258.604	912.250	912.190	0.222	EARTHEN TRAPEZOIDAL	300	
151	81T	81i	D88aT	5.72	E=264104.264 N=2506258.604	E=264101.546 N=2506262.864	912.190	912.180	0.210	EARTHEN TRAPEZOIDAL	300	TRANSITION
152	81i	67i	D89	15.21	E=264101.546 N=2506262.864	E=264113.879 N=2506270.570	912.180	912.100	0.526	R.C. PIPE DRAIN	ø700	
153	82	83	D90	85.62	E=264128.527 N=2506243.174	E=264201.591 N=2506278.216	912.250	912.050	0.232	EARTHEN TRAPEZOIDAL	400	
154	67i	84	D90a	143.77	E=264113.879 N=2506270.570	E=264222.684 N=2506291.397	912.100	911.900	0.139	EARTHEN TRAPEZOIDAL	800	
155	83	84	D91	24.87	E=264201.591 N=2506278.216	E=264222.684 N=2506291.397	912.050	911.980	0.286	R.C. PIPE DRAIN	ø500	
156	84	85	D92	24.04	E=264222.515 N=2506291.291	E=264242.827 N=2506303.983	911.900	911.760	0.603	EARTHEN TRAPEZOIDAL	800	
157	85	86	D93	12.63	E=264242.827 N=2506303.983	E=264249.398 N=2506293.468	911.760	911.690	0.515	R.C. PIPE DRAIN	ø700	
158	86	87	D94	202.25	E=264249.398 N=2506293.468	E=264290.109 N=2506120.188	911.690	910.860	0.413	EARTHEN TRAPEZOIDAL	800	
159	87	87T	D95	171.47	E=264290.109 N=2506120.188	E=264380.941 N=2505974.825	910.860	910.520	0.197	EARTHEN TRAPEZOIDAL	800	
160	87T	88	D95T	5.26	E=264380.941 N=2505974.825	E=264383.627 N=2505970.527	910.520	910.410	2.033	EARTHEN TRAPEZOIDAL	800	TRANSITION
161	53	53T	D96	22.58	E=264360.136 N=2505955.848	E=264379.421 N=2505967.899	910.950	910.500	2.002	EARTHEN TRAPEZOIDAL	1000	
162	53T	88	D96T	4.99	E=264379.421 N=2505967.899	E=264383.627 N=2505970.527	910.500	910.410	1.762	EARTHEN TRAPEZOIDAL	1000	TRANSITION
163	88	89	D97	36.61	E=264383.627 N=2505970.527	E=264417.672 N=2505957.370	910.410	910.230	0.492	EARTHEN TRAPEZOIDAL	1500	TO STORM WATER POND
164	101	102	D101	18.8	E=263833.462 N=2505844.775	E=263843.424 N=2505828.832	912.700	912.605	0.500	R.C. PIPE DRAIN	ø500	TO NATURAL OUTFALL
165	103	104	D102	30.6	E=264061.269 N=2505655.895	E=264077.485 N=2505629.945	911.800	911.647	0.500	R.C. PIPE DRAIN	ø500	TO NATURAL OUTFALL
166	105	106	D103	17.3	E=264495.568 N=2505936.548	E=264512.780 N=2505934.807	910.200	910.113	0.500	R.C. PIPE DRAIN	ø700	OVERFLOW OUTFALL
167	107	108	D104	30.6	E=264517.933 N=2506157.587	E=264548.377 N=2506154.502	910.700	910.547	0.500	R.C. PIPE DRAIN	ø600	TO NATURAL OUTFALL
168	109	110	D105	30.6	E=264190.394 N=2506450.067	E=264216.344 N=2506466.282	912.200	912.047	0.500	R.C. PIPE DRAIN	ø500	TO NATURAL OUTFALL
169	111	112	D106	30.6	E=264138.734 N=2506532.740	E=264164.684 N=2506548.956	912.350	912.197	0.500	R.C. PIPE DRAIN	ø700	TO NATURAL OUTFALL
170	113	114	D107	30.6	E=264505.212 N=2506031.861	E=264535.656 N=2506028.776	910.600	910.447	0.500	R.C. PIPE DRAIN	ø700	TO NATURAL OUTFALL

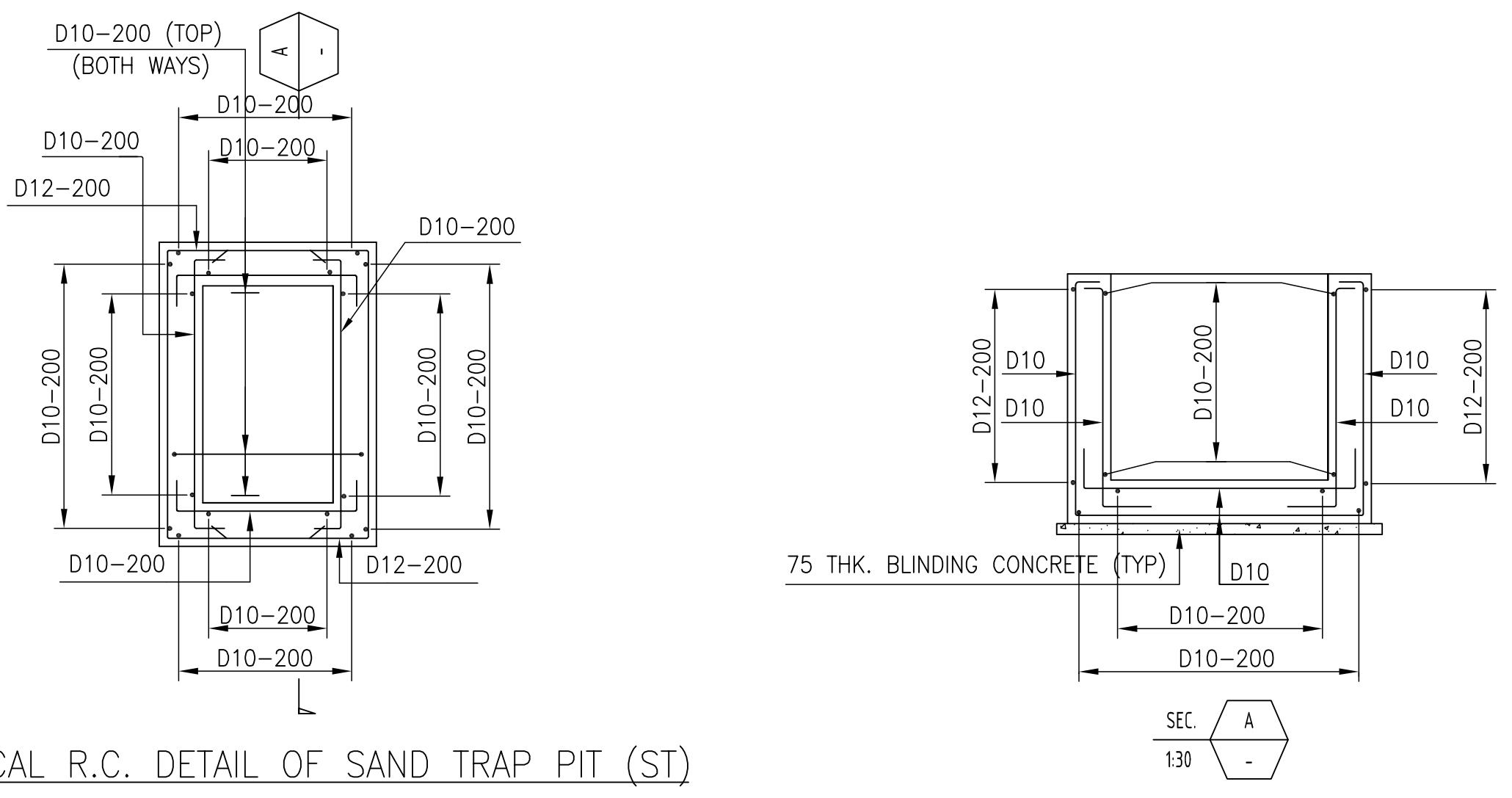


CL. OF EARTHEN DRAIN
CL. OF EARTHEN DRAIN
CENTER CO-ORDINATE OF SAND TRAP PIT
WIDTH (W)
LENGTH (L)
200
200
75 THK. P.C.C M15 LYNING WITH SKIN REINFORCEMENT (6MM DIA 150 SQUARE MESH)
BERM
WIDTH OF ROAD
BERM

PLAN
TYPICAL G.A. DETAIL OF SAND TRAP PIT (ST)
SCALE= 1:50



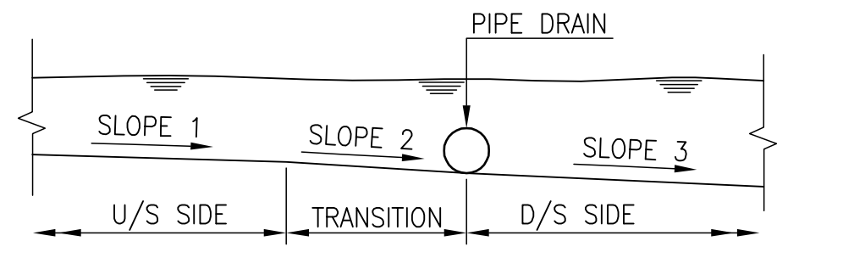
TYPICAL CROSS SECTION OF ROAD WITH EARTHEN DRAIN AND SAND TRAP PIT
SCALE= 1:50



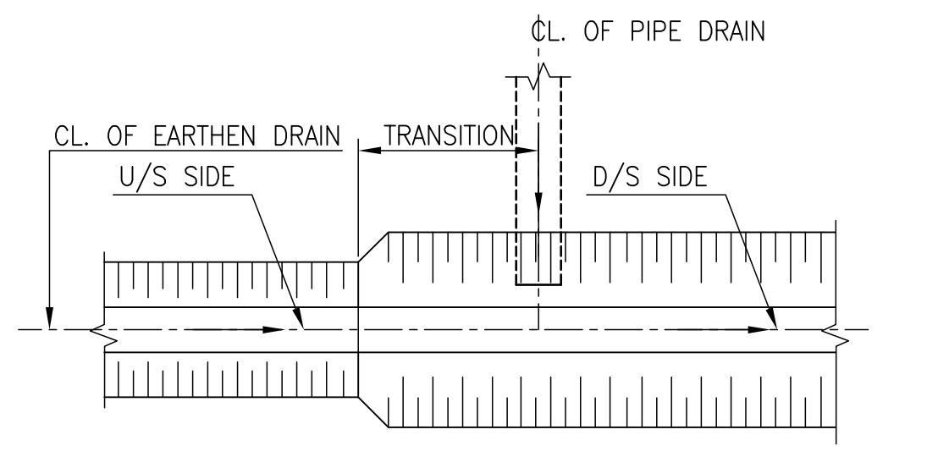
TYPICAL R.C. DETAIL OF SAND TRAP PIT (ST)

SCHEDULE OF SAND TRAP PIT

SL. NO.	PIT MKD.	SIZE OP PIT (MM)	CO-ORDINATES AT CENTER OF PIT		REMARKS
			E(M)	N(M)	
1	ST1	500 (W) x 1000 (L)	E=263771.667	N=2506257.736	
2	ST2	500 (W) x 1000 (L)	E=263815.824	N=2506187.069	
3	ST3	500 (W) x 1000 (L)	E=263795.916	N=2506055.459	
4	ST4	500 (W) x 1000 (L)	E=263877.197	N=2506088.851	
5	ST5	500 (W) x 1000 (L)	E=263851.552	N=2506045.352	
6	ST6	500 (W) x 1000 (L)	E=263933.214	N=2505990.715	
7	ST7	500 (W) x 1000 (L)	E=263900.506	N=2505930.064	
8	ST8	800 (W) x 1200 (L)	E=263996.347	N=2505898.172	
9	ST9	500 (W) x 1000 (L)	E=263972.712	N=2505813.324	
10	ST10	800 (W) x 1200 (L)	E=264120.356	N=2505820.639	
11	ST11	500 (W) x 1000 (L)	E=264038.122	N=2505856.607	
12	ST12	500 (W) x 1000 (L)	E=264079.478	N=2506000.745	
13	ST13	500 (W) x 1000 (L)	E=264128.756	N=2505968.154	
14	ST14	500 (W) x 1000 (L)	E=264183.305	N=2505880.858	
15	ST15	600 (W) x 1200 (L)	E=264263.472	N=2505895.446	
16	ST16	500 (W) x 1000 (L)	E=263996.714	N=2506100.208	
17	ST17	500 (W) x 1000 (L)	E=264066.859	N=2506035.722	
18	ST18	500 (W) x 1000 (L)	E=263969.919	N=2505965.753	
19	ST19	500 (W) x 1000 (L)	E=264072.907	N=2506011.260	
20	ST20	500 (W) x 1000 (L)	E=264146.264	N=2506085.340	
21	ST21	500 (W) x 1000 (L)	E=264209.966	N=2506008.028	
22	ST22	500 (W) x 1000 (L)	E=264128.632	N=2505991.753	
23	ST23	500 (W) x 1000 (L)	E=264191.074	N=2505982.745	
24	ST24	500 (W) x 1000 (L)	E=264261.974	N=2505924.799	
25	ST25	800 (W) x 1200 (L)	E=264231.640	N=2505890.177	
26	ST26	500 (W) x 1000 (L)	E=264258.885	N=2506035.420	
27	ST27	500 (W) x 1000 (L)	E=264221.646	N=2506012.737	
28	ST28	500 (W) x 1000 (L)	E=264292.662	N=2505928.307	
29	ST29	500 (W) x 1000 (L)	E=264309.896	N=2505953.783	
30	ST30	1000 (W) x 1500 (L)	E=264339.004	N=2505942.643	
31	ST31	500 (W) x 1000 (L)	E=264119.016	N=2506106.167	
32	ST32	500 (W) x 1000 (L)	E=263999.610	N=2506118.973	
33	ST33	500 (W) x 1000 (L)	E=264041.814	N=2506051.434	
34	ST34	500 (W) x 1000 (L)	E=264118.129	N=2506082.381	
35	ST35	500 (W) x 1000 (L)	E=264109.883	N=2506144.182	
36	ST36	500 (W) x 1000 (L)	E=264233.686	N=2506099.147	
37	ST37	500 (W) x 1000 (L)	E=264286.661	N=2506014.368	
38	ST38	500 (W) x 1000 (L)	E=264344.110	N=2505960.455	
39	ST39	500 (W) x 1000 (L)	E=264193.690	N=2506258.657	
40	ST40	500 (W) x 1000 (L)	E=264253.838	N=2506234.090	
41	ST41	500 (W) x 1000 (L)	E=264283.950	N=2506106.644	
42	ST42	500 (W) x 1000 (L)	E=264351.198	N=2505999.024	
43	ST43	500 (W) x 1000 (L)	E=263827.188	N=2506194.170	
44	ST44	500 (W) x 1000 (L)	E=263941.567	N=2506194.151	
45	ST45	500 (W) x 1000 (L)	E=264046.681	N=2506259.833	
46	ST46	800 (W) x 1200 (L)	E=264151.330	N=2506271.100	
47	ST47	500 (W) x 1000 (L)	E=263976.134	N=2506201.129	
48	ST48	500 (W) x 1000 (L)	E=264053.144	N=2506249.250	
49	ST49	500 (W) x 1000 (L)	E=264115.405	N=2506240.773	
50	ST50	500 (W) x 1000 (L)	E=263924.679	N=2506115.795	
51	ST51	500 (W) x 1000 (L)	E=264083.377	N=2506204.348	
52	ST52	800 (W) x 1200 (L)	E=264235.812	N=2506299.600	
53	ST53	800 (W) x 1200 (L)	E=264278.288	N=2506218.362	
54	ST54	800 (W) x 1200 (L)	E=264294.466	N=2506113.215	
55	ST55	800 (W) x 1200 (L)	E=264361.714	N=2506005.595	
56	ST56	1500 (W) x 2500 (L)	E=264405.342	N=2505957.954	



ELEVATION VIEW OF TRANSITION



TYPICAL DETAIL OF TRANSITION
SCALE= 1:50

(LOCATION OF TRANSITIONS, LEVELS, SLOPE AND ALL DIMENSIONS REFER SCHEDULE OF STORM WATER DRAINAGE SYSTEM AND RESPECTIVE LAYOUT DRAWING)

FOR ADDITIONAL ROAD CROSSING DRAINAGE PIPE AT MOBILE CRANE AND VEHICLE OPERATIONAL AREA, REFER DRAWING NO. MD-216-49EA-EG-CV-GAD-1002 (SH 3).

THIS DRAWING SHOULD BE READ IN CONJUNCTION WITH DRAWING NO.
MD-216-49EA-EG-CV-DED-1002
MD-216-49EA-EG-CV-GAD-1001 &
MD-216-49EA-EG-CV-GAD-1002

LEGEND:

DRAIN.	DRAINAGE
DET.	DETAILS
LAY.	LAYOUT
U.N.O	UNLESS NOTED OTHERWISE
R.C.	REINFORCED CONCRETE
NET.	NETWORK
MKD.	MARKED
P.C.C	PLAIN CEMENT CONCRETE
CL.	CENTRE LINE
DIA	DIAMETER
C/C	CENTRE TO CENTRE
FGL	FINISHED GROUND LEVEL
U/S	UPSTREAM
D/S	DOWNSTREAM
SH	SHEET

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETRES
2. CO-ORDINATES AND LEVELS ARE IN METRES
3. CO-ORDINATES ARE ON WORLD GRID (FLAT PLANT/WORLD)
4. WORLD MAPPING GRID BASED ON UTM (WGS 84) Q38N
5. GRADE OF REINFORCED CEMENT CONCRETE SHALL BE M35.
6. REINFORCEMENT BARS SHALL CONFORM TO ASTM A615 GRADE 60 (fy=420 N/sq.mm.)
7. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE 25mm(MINIMUM) U.N.O.
8. DRAIN SECTIONS TO BE LINED WITH 50-75 THICK P.C.C IN CASE OF UNSTABLE OR SOFT SOIL. FOR STIFF SOIL AND ROCK NO LINER IS REQUIRED.

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		D01	08/09/2023	AS-BUILT	APEX	BJB	APEX	SJM	
		C02	02/06/2021	ISSUED FOR CONSTRUCTION & REVISED AS MARKED C02	APEX	BJB	APEX	SJM	
		C01	20/02/2021	ISSUED FOR CONSTRUCTION	APEX	BJB	APEX	SJM	
		A05	25/01/2021	REVISED AS PER EMPLOYER'S COMMENT ON 08/01/2021 & AS MKD. A05	APEX	BJB	APEX	SJM	
		A04	11/12/2020	REVISED AS PER EMPLOYER'S COMMENT ON 10/11/2020	APEX	SNM	APEX	SJM	
				& ISSUED FOR REVIEW MARKED A04					
		A03	14/10/2020	REVISED AS PER EMPLOYER'S COMMENTS ON 16/08/2020 & AS MKD. A03	SB	SNM	APEX	SJM	
		A02	29/07/2020	GENERALLY REVISED AS PER EMPLOYER'S COMMENTS ON 30/04/2020	SB	SNM	APEX	SJM	
MD-216-49EA-EG-CV-GAD-1002	LAY. & DET. OF STORM WATER DRAINAGE NET. SH 1 & SH 2			& AS MARKED A02					
MD-216-49EA-EG-CV-GAD-1001	LAY. AND DETAILS OF STORM WATER DRAINAGE NET.	A01	31/03/2020	ISSUED FOR REFERENCE	SB	SNM	APEX	SJM	

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP

RESERVER FOR VENDOR LOGO AND DRAWING NUMBER

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Saudi Arabian Mining Company شركة التعدين العربية السعودية

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LARSEN & TOUBRO DRAWING NUMBER	019045_910_C_V_24019_001		REVISION 7
OUTOTEC DRAWING NUMBER	900707 __ GAU01 __ AG90 __ CTB17 __ 00001	DOCUMENT ID 0U100673472	REVISION 7

DRAWN	SB	27/03/2020
CHECKED	SNM	29/03/2020
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APPROVED	SJM	31/03/2020
EMPLOYER		
SCALE	1:30, 1:50, 1:1500	A0

MANSOURAH - MASSARAH GOLD PROJECT
DETAIL DRAWING
PROCESS PLANT STORM WATER DRAINAGE NET.
LAY. & DET. OF STORM WATER DRAIN. NET.-SH 3

DRAWING NUMBER: MD - 216 - 49EA - EG - CV - DED - 1001
REV: D01